A New Grading System of Lumbar Central Canal Stenosis on MRI: An Easy and Reliable Method

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Inventory Category:	Clinical grading / reference standard
Comparability / Priority:	Very High | Priority: Critical

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Clinical grading standard | Axial lumbar MRI
Dataset & Cohort:	nan (Sample: nan)
Disease / Target:	Central canal stenosis
Model / AI Method:	Manual central-canal grading
Evaluation Metrics:	Agreement
Key Finding / Relevance:	nan
Thesis Context (Ch 2):	Clinical severity labels used by later AI stenosis systems need to be grounded in such grading standards.

Abstract / Summary

ObjectiveTo introduce a new grading system of lumbar central canal stenosis, evaluate its reliabilities, and compare it to the cross-sectional area and anterior-posterior diameter of the dural sac.Materials and methodsLumbar central canal stenosis is defined as obliteration of the anterior CSF space in front of the cauda equina. Four musculoskeletal radiologists independently graded lumbar central canal stenosis by this new grading system based on separation degree of the cauda equina on T2-weighted axial images (grade 0 = no lumbar stenosis without obliteration of anterior CSF space; grade 1 = mild stenosis with separation of all cauda equina; grade 2 = moderate stenosis with some cauda equina aggregated; and grade 3 = severe stenosis with none of the cauda equina separated) in 81 patients to determine inter- and intra-reader reliability. One radiologist measured cross-sectional areas and anterior-posterior diameters and compared these to lumbar central canal stenosis grades.ResultsInter-reader reliabilities were substantial to almost perfect (ICC reliability = 0.730-0.953). Intra-reader reliability was almost perfect (kappa value = 0.863-0.900). Cross-sectional areas and anterior-posterior diameters were different according to grades at all levels ($p = 0.000$-0.049), except between grades 2 and 3 of L2-3. At L5-S1, only anterior-posterior diameter was different between grades 0 and 1 ($p = 0.005$) and between grades 0 and 2 ($p = 0.022$).ConclusionsThis new grading system may be helpful to clinicians for simple and practical evaluation of lumbar central canal stenosis and for communicating with each other.